

Post-Lock Geometry — The Initial Plane and the Terminal Plane Span 4 Dimensions (Two Nonzero Principal Angles); Plane Switching Is Not an Aggregation Artifact of a Long-Window SVD

Noriaki Kihara (WF System Co., Ltd.), ORCID 0009-0004-6753-4020

2026-09-12

Series: Foundations of Self-Consistent Relational-Wave Closed Systems (Paper 7 of 10)

Author: Noriaki Kihara (WF System Co., Ltd.) **Date:** 2026-09-12

Version DOI: 10.5281/zenodo.22729108

Concept DOI: 10.5281/zenodo.22729107

ORCID: 0009-0004-6753-4020

Zenodo: <https://zenodo.org/records/22729108>

Abstract

We fix geometrically what inflation (Paper 6) and SSB lock (Paper 5) do in state space. The map is a real orthogonal rotation $z(t) = Q_t z_0$, and the state remains on a 2-dimensional plane (rank 2) at every instant (the singular values of $[\Re z, \Im z]$ take only two values at every step, and $z^T z = 0$ conservation gives an orthogonal isometric frame). On the floor the floor plane $P_0 = \text{span}\{\Re z_0, \Im z_0\}$ is K -invariant and the state stays in P_0 . When SSB leaves the floor, the lock plane P_{term} tilts away from P_0 , and the principal angles between the two planes are, independently of N , **both nonzero** ($N = 6:27.1^\circ, 30.8^\circ \text{ } \diagup \text{ } N = 40:18.1^\circ, 20.4^\circ$), so the dimension spanned by the two planes is 4. P_0 and P_{term} span 4 dimensions with two nonzero principal angles, and from the same floor the terminal orientation is seed-dependent and not uniquely determined. We do not claim a specific $SO(4)$ rotation structure for the cumulative map (in the real data M_t is confirmed not to preserve a 4-dimensional subspace). The tilt of the lock plane away from the floor is, over all $N = 6\text{--}40$, $14.2^\circ\text{--}29.0^\circ$ (median $\sim 19.2^\circ$). During the transition a second direction rises (transient $s_3/s_1 \approx 0.10\text{--}0.20 = \text{effective rank } 4$), and after lock the system re-locks to a single plane (tail $s_3/s_1 \approx 10^{-5}\text{--}10^{-13} = \text{rank } 2$). By a short-window SVD test we separate this “plane switching” from an appearance of long-window aggregation, showing it is a **real plane tilt**. Classification: derived geometry + measurement (artifact excluded).

1. Problem

Reading the endpoint merely as a “circle” fails to capture the essential observation of this map — **switching from one plane to another plane**. We quantify this for all $N \geq 6$ and separate the appearance (aggregation of a long-window SVD) from the real rotation.

2. Instantaneous Structure — Rank 2 (a Single Plane)

The map is a real orthogonal rotation $z(t) = Q_t z_0$ (Q_t real orthogonal, acting identically on $\Re z, \Im z$), and the state stays on a 2-dimensional plane at every instant (the singular values of $[\Re z, \Im z] \in \mathbb{R}^{2 \times M}$ take only two values at every step, and $z^\top z = 0$ conservation gives an orthogonal isometric frame). **Formulation as Grassmann motion:** for $z = a + ib$, from $z^\top z = 0$ and $\|z\| = 1$ we have $\|a\|^2 = \|b\|^2 = \frac{1}{2}$ and $a^\top b = 0$, so $u = \sqrt{2}a$, $v = \sqrt{2}b$ are always an orthonormal real 2-frame $(u, v) \in V_2(\mathbb{R}^M)$. The global phase $z \mapsto e^{i\theta}z$ merely rotates (u, v) within the same plane, so the object observed — modulo global phase — is the **2-dimensional subspace** $P(t) = \text{span}\{u, v\}$ **itself** (motion on the Grassmann manifold $\text{Gr}(2, M)$). The projection $\Pi_t = u_t u_t^\top + v_t v_t^\top$ moves exactly, by each step’s real orthogonal $Q_t = e^{\Delta\tau K_t}$, as

$$\Pi_{t+1} = Q_t \Pi_t Q_t^\top$$

On the floor $Q_0 P_0 = P_0$, so $\Pi_{t+1} = \Pi_t$ (relative equilibrium, fixed); in inflation $\Pi_t \neq \Pi_0$ and the plane moves; at saturation it re-locks to P_{term} . Sections 3–5 measure this plane motion $\Pi_0 \rightarrow \Pi_t \rightarrow \Pi_{\text{term}}$. On the floor $P_0 = \text{span}\{\Re z_0, \Im z_0\}$ is K -invariant ($Ka = \mu b$, $Kb = -\mu a$) and the state stays in P_0 (principal angles zero). The state-space SVD ratio s_3/s_1 of the third/first singular values over the last 200 steps after lock is 10^{-5} – 10^{-13} = a single plane (rank 2/ S^1).

3. The Dimension Spanned by the Initial Plane and the Terminal Plane — 4D Span

When SSB leaves the floor, the lock plane P_{term} tilts away from P_0 . The principal angles between the initial plane P_0 and the terminal plane P_{term} are, independently of N , **both nonzero** ($N = 6:27.1^\circ, 30.8^\circ$ / $N = 40:18.1^\circ, 20.4^\circ$), so the dimension spanned by the two planes is 4. The tilt of the lock plane away from the floor ($\arcsin \sqrt{H_\perp/H_{\text{end}}}$) is, over all $N = 6$ – 40 , as follows (excerpted from `plane_switch_allN.csv`):

N	tilt $[\circ]$	transient s_3/s_1	tail s_3/s_1
6	29.0	0.172	3.8×10^{-6}
8	20.3	0.135	2.8×10^{-4}
16	24.5	0.172	1.1×10^{-5}
22	15.7	0.130	1.1×10^{-13}
40	19.3	0.132	4.5×10^{-7}
64	21.7	0.021	2.5×10^{-4}

Over all N the tilt is 14.2° – 29.0° (median $\sim 19.2^\circ$), neither 0° nor 90° = isotropy is broken.

The tilt, the two principal angles, and $H_\perp/H$ are not independent measured quantities but the same Grassmann geometry. For the initial and terminal frames $U_0 = (u_0, v_0)$, $U_1 = (u_1, v_1)$ and $P_0^\perp = I - U_0 U_0^\top$, from the definition of principal angles (the singular values of $U_0^\top U_1$ equal $\cos \theta_i$) we have $\|P_0^\perp U_1\|_F^2 = \sin^2 \theta_1 + \sin^2 \theta_2$. Since $z = \frac{1}{\sqrt{2}}(u + iv)$,

$$\frac{H_\perp}{H} = \frac{1}{2} (\sin^2 \theta_1 + \sin^2 \theta_2) = \frac{1}{4} \|\Pi_{\text{term}} - \Pi_0\|_F^2, \quad \alpha = \arcsin \sqrt{\frac{\sin^2 \theta_1 + \sin^2 \theta_2}{2}}$$

(from $\Pi = U U^\top$ we get $\|\Pi_1 - \Pi_0\|_F^2 = 2(\sin^2 \theta_1 + \sin^2 \theta_2)$). This is not an additional hypothesis but an **exact geometric identity** from the definitions, and $N = 6$'s ($27.1^\circ, 30.8^\circ$) $\rightarrow \alpha = 29.0^\circ$ and $N = 40$'s ($18.1^\circ, 20.4^\circ$) $\rightarrow \alpha = 19.3^\circ$ agree with the table. That is, the tilt α away from the floor is the **Grassmann distance** $\frac{1}{4} \|\Pi_{\text{term}} - \Pi_0\|_F^2$ from the initial plane itself.

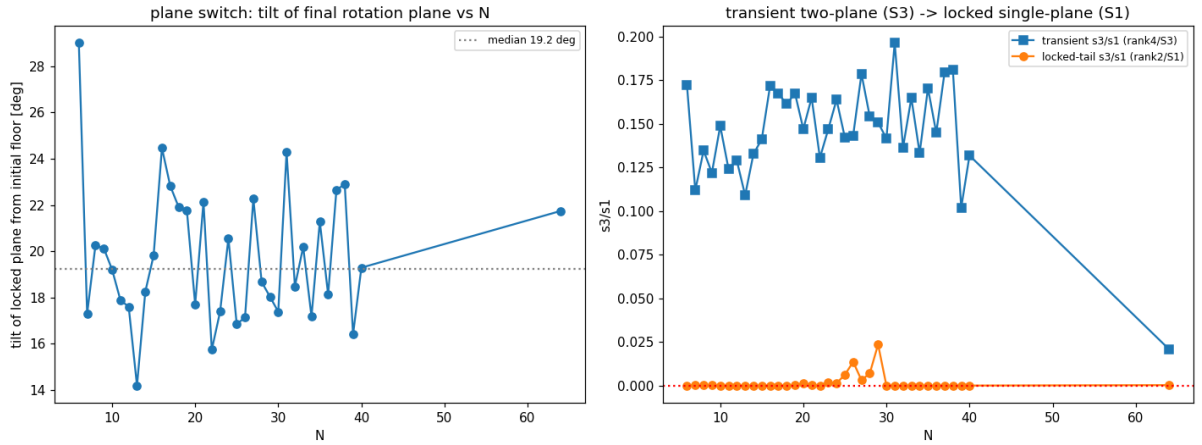


Figure 1: Plane switching (all $N \geq 6$). Left = tilt of the lock plane away from the floor vs N ; right = transient s_3/s_1 (rank 4/ S^3) $\rightarrow$ tail s_3/s_1 (rank 2/ S^1).

Figure 1 N -dependence of plane switching.

4. The Initial Plane and the Terminal Plane Span 4 Dimensions

P_0 and P_{term} meet at two nonzero principal angles ($N = 6:27.1^\circ, 30.8^\circ \nearrow N = 40:18.1^\circ, 20.4^\circ$) and $\dim(P_0 + P_{\text{term}}) = 4$ — the terminal plane tilts away from the initial plane in two independent directions (these “4 directions” are meant only in the sense of the dimension of the span). **From the same floor this terminal orientation is seed-dependent and not uniquely determined** (Paper 5). We do not claim a specific $SO(4)$ rotation structure for the cumulative map $M_t = \prod_n \exp(\Delta \tau K(\varphi_n))$ — in the real data M_t is confirmed not to preserve this 4-dimensional subspace (the invariance $M_t S = S$ is broken), and M_t is an orthogonal rotation of the full $\mathbb{R}^M$ that tilts P_0 into the 4-dimensional span.

During the transition a second rotation plane rises (transient $s_3/s_1 \approx 0.10\text{--}0.20 = \text{rank } 4$), but after lock the system re-locks to a single plane (tail $s_3/s_1 \approx 0 = \text{rank } 2$). At $N = 64$ the transient $s_3/s_1 = 0.021$ is small, so the second-plane transient weakens for large N . This transitional rank 4 is temporary; after lock it returns to rank 2, and the cumulative map also does not preserve a 4-dimensional subspace (§4) — **no persistent S^3 structure is found** (inspected, short-window SVD + discriminating computation).

In the time-varying, non-normal tangent cocycle confirmed in Paper 6, the maximally amplified direction rotates with time while it is amplified. The additional direction of the transient observed in this paper can be read as the geometric manifestation of that tangent dynamics tilting the instantaneous rank-2 plane $P(t)$ with time. Therefore the effective rank 4 of the transient is **not a 4-dimensionalization of the instantaneous state but the local span swept in a short time by the moving 2-plane $P(t)$** , and after saturation it re-locks at rank 2 to a single terminal plane. That is, Paper 6 (in tangent space the amplified direction rotates) and Paper 7 (in state space the 2-plane $P(t)$ tilts and moves to another 2-plane) are the tangent-side and state-side representations of the same phenomenon. Using $H_{\perp}/H = \frac{1}{4}\|\Pi_t - \Pi_0\|_F^2$ (the §3 identity), Paper 6’s initial ramp $H_{\perp}/H \propto e^{2\gamma t}$ is geometrically $\|\Pi_t - \Pi_0\|_F \propto e^{\gamma t}$ — **inflation is an exponential departure from the initial plane on the Grassmann space $\text{Gr}(2, M)$** .

5. Exclusion of the Artifact

A long-window SVD can, if a single plane rotates with time, be aggregated into an apparently higher rank. With the short-window SVD test (`artifact_test_shortwindow_svd`) we separated the fact that, instantaneously staying at rank 2, the initial plane P_0 tilts in two directions (P_0 and P_{term} span 4 dimensions) — the “plane switching” is not an appearance of long-window aggregation but a **real plane tilt**.

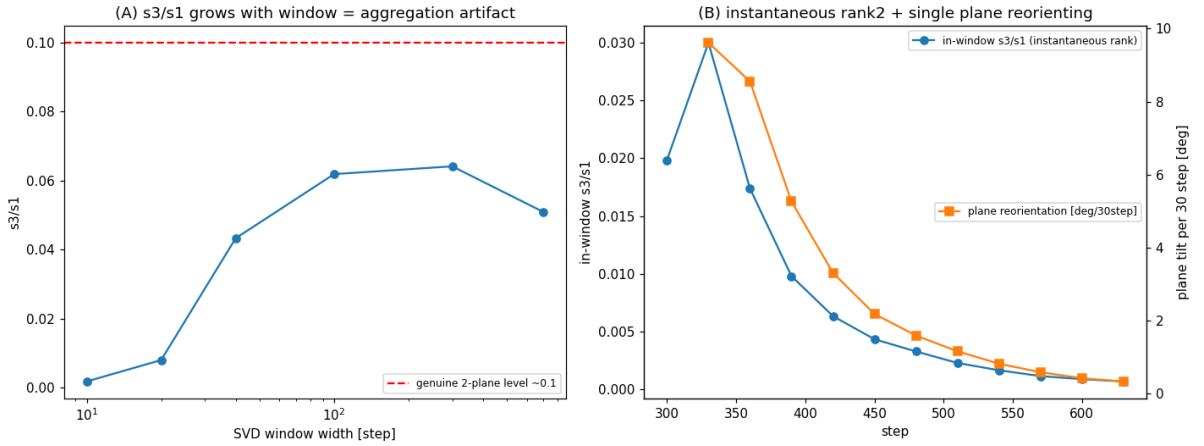


Figure 2: Short-window SVD test. While preserving instantaneous rank 2, the initial plane tilts in two directions (plane switching is real).

Figure 2 Artifact exclusion (short-window SVD).

6. A Caveat on Readout

When there is initially only a single complex plane, its normal is a degenerate orthogonal complement and is unobservable (conceptual). Only once inflation raises a specific direction in the normal space does the relative phase (tilt) between the initial plane and the new plane become readable. **What can be read out is always a relational quantity (the relative phase between planes)**; the absolute normal direction cannot be recovered from the state alone (Q_t hides the orientation, and there is no conserved quantity pointing to it).

7. Claim Classification and Open Items

- Classification: derived geometry (real orthogonal rotation rank 2) + measurement (principal angles, tilt, 4D span, artifact exclusion) + discriminating computation (confirmed that the cumulative map does not preserve a 4-dimensional subspace and is not $SO(4)$). Verdict: the 4D span, the two nonzero principal angles, and the double rotation of the relative configuration are retained; “the dynamics being an $SO(4)$ double rotation on 4 dimensions” is rejected.
- Reference (a different regime): the crystalline terminal states at $N = 3, 4$ are special solutions in which the plane refreezes at the magic angle $\arccos \sqrt{2/3} = 35.264^\circ$ (the general $N \geq 6$ lock is $14\text{--}29^\circ$). The crystal/glass bifurcation is a separate topic of this series.
- Open: the physical identification of the 4D span ($xyzt$ or $xyztRQ$, etc.) and the type of gauge (scale) are outside the scope of this paper (Paper 0 §Open).

Related Work (Structural Agreement, Not the Source of Derivation)

- The double rotation of $SO(4)$ (two invariant planes, two independent rotation angles): this is a description of the **relative configuration** of $P_0 \rightarrow P_{\text{term}}$, not a property of the cumulative map itself (in the §4 discriminating computation $M_t S \neq S$).
- Relative equilibrium (Marsden & Ratiu): the rigid-body-rotation relative equilibrium of the floor and the lock.

References

1. Noriaki Kihara, “The mechanism of inflationary rapid expansion in self-consistent relational-wave closed systems,” Concept DOI 10.5281/zenodo.22112008.
2. J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, Springer (1999).

Reproduction

The analysis scripts, figures, and CSV are in 位相ロック全 N 確認_相対平衡_20260912/面の乗り換え全 N_20260912/ (analyze_plane_switch_allN_20260912.py, artifact_test_shortwindow_svd_20260912.py, plane_switch_allN.csv, fig_plane_switch_allN.png, fig_artifact_shortwindow_svd.png). For the locked data, small N and $N = 23\text{--}29$ are 500 steps, and the unsaturated N (22, 30–40) and $N = 64$ are 2000 steps. The principal angles of the initial and terminal planes are the step0/step2000 complex planes in N22_2000step 検証走行_/ and N40_2000step 検証走行_/. The early rank-3 audit is in independent_rank3_geometry_audit_20260831/. The dynamics map is identical to the canonical run_N3_N40_stage123_v1.py (SHA256 1abf2353fee2e4f56f05e7a6f149fd086885136beb61ab571b48a56b09691567), readout only.